

**RED SEAL EXAM PREP**

# **456A Welder**

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings  
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## Mock Exam — answer all questions, then check the key

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**Q1.** Which filler metal is used for GTAW of chrome-moly (Cr-Mo) steel such as P91?

- A) ER308L
- B) ER70S-2
- C) ER90S-B9
- D) ER4043

**Q2.** During GMAW the arc pops and stutters, the wire occasionally burns back to the contact tip, and the bead comes out irregular. What is the most likely cause?

- A) Work angle too steep for the joint
- B) Worn drive rolls or a kinked liner
- C) Travel speed too slow for the amperage
- D) Shielding gas flow rate set too low

**Q3.** What is the main characteristic of an E6010 electrode?

- A) Rutile coating, easy slag removal, AC or DC
- B) High cellulosic coating, deep penetration, DCEP only
- C) Iron powder coating, high deposition rate
- D) Low hydrogen, requires dry storage

**Q4.** What does "stringer bead" mean in multi-pass welding?

- A) A bead deposited in the vertical-up position
- B) A narrow bead deposited with no side-to-side motion
- C) A wide weave bead deposited side to side
- D) The final cap pass over a multi-pass weld

**Q5.** A welder notices the electrode holder cable insulation is cracked. The correct action is:

- A) Continue welding if the crack is small
- B) Replace the cable before continuing work
- C) Wrap the crack with electrical tape and continue
- D) Reduce amperage to lower risk

**Q6.** What is the Heat Affected Zone (HAZ) in welding?

- A) The area heated by the torch before welding starts
- B) The area of the weld covered by the shielding gas
- C) Base metal altered by welding heat but not melted
- D) The weld metal itself, meaning the fusion zone

**Q7. What is the function of anti-spatter spray in GMAW?**

- A) To reduce oxidation of the weld pool
- B) To lubricate the wire liner
- C) To improve shielding gas coverage
- D) To keep spatter from sticking to the nozzle

**Q8. E6013 is preferred for welding thin sheet metal because:**

- A) It provides maximum penetration into thin material
- B) It produces a hard slag that prevents porosity
- C) It has a soft arc, easy slag removal, and works on AC or DC
- D) It generates the highest amperage range

**Q9. Excessive spatter in FCAW-G is most likely caused by:**

- A) Wire stickout too short
- B) Shielding gas flow rate too high
- C) Arc voltage set too low
- D) Base metal temperature too high

**Q10. Which GMAW transfer mode requires the highest arc voltage to achieve?**

- A) Short circuit transfer
- B) Pulsed spray transfer
- C) Globular transfer
- D) Spray transfer

**Q11. In the FCAW electrode classification E71T-1, what do the "T" and the trailing "-1" designate?**

- A) T = travel direction; -1 = flat position only
- B) T = tubular electrode; -1 = single-pass use only
- C) T = titanium flux coating; -1 = wire diameter
- D) T = tubular electrode; -1 = CO<sub>2</sub> or Ar/CO<sub>2</sub>, DCEP

**Q12. A 456A welder is welding mild steel at a fixed indoor workstation. Under CSA W117.2 and provincial occupational health and safety regulation, what determines whether the ventilation is adequate?**

- A) 1,000 cfm of general dilution air per welder whenever the shop ceiling is under 5 m
- B) Nothing below 100 A — ventilation applies only to high-current welding on coated metal
- C) Fume kept below the occupational exposure limits, with local exhaust at the source
- D) 2,000 cfm of general dilution air per welder, the fixed minimum set by Canadian regulation

**Q13. What is the purpose of a Charpy V-notch impact test?**

- A) To measure the toughness of the weld at a set temperature
- B) To measure the hardness of the HAZ
- C) To measure the tensile strength of the weld metal
- D) To measure porosity in the weld cross-section

**Q14.** In oxyfuel cutting (OFC), what initiates the cutting action?

- A) A high-pressure jet of pure oxygen
- B) An electric arc combined with oxygen
- C) The heat from the preheat flames alone
- D) The acetylene flame melting the metal

**Q15.** ER70S-6 wire has higher silicon and manganese content than ER70S-3. Why?

- A) To allow welding with 100% CO<sub>2</sub> shielding gas only
- B) To enable AC power source compatibility
- C) To add deoxidizers for welding slightly rusty surfaces
- D) To increase tensile strength above 80 ksi

**Q16.** What causes excessive drag lines (curved) on an OFC cut face?

- A) Cutting speed too slow
- B) Oxygen pressure too high
- C) Cutting speed too fast
- D) Preheat flames too large

**Q17.** What is globular transfer in GMAW and when does it occur?

- A) Transfer of large droplets at intermediate voltage — high spatter
- B) Transfer of very fine droplets at high current — preferred for production welding
- C) Transfer where the wire dips into the molten pool at low voltage
- D) Transfer mode unique to pulsed GMAW at low background current

**Q18.** After chipping the slag off a flux-cored fillet, the welder finds narrow elongated grooves running along the crown of the bead. What are they?

- A) Undercut, where the arc melted the toe faster than the puddle refilled it
- B) Arc strikes, where the wire touched the plate outside the joint on restart
- C) Worm tracks, from gas trapped between the weld metal and freezing slag
- D) Slag inclusions, where flux rolled ahead of the arc and was welded over

**Q19.** What does "DCEP" mean in SMAW?

- A) Direct Current Electrode Positive — electrode is negative, work is positive
- B) Direct Current Electrode Polarity — a setting found only on AC machines
- C) Direct Current Electrode Positive — electrode is positive, work is negative
- D) Dual Current Electrode Positive — a machine setting allowing both AC and DC

**Q20.** What is the purpose of the high-frequency (HF) start in GTAW?

- A) To generate the shielding gas ionization
- B) To start the arc without touching the workpiece
- C) To increase amperage for thicker material
- D) To provide the cleaning action for aluminum

**Q21.** Low-hydrogen electrodes (E7018) must be stored in:

- A) A heated rod oven at 120–150°C (250–300°F)
- B) A freezer to prevent oxidation
- C) Any dry location away from direct sunlight
- D) A sealed plastic bag at room temperature

**Q22.** What does "walking the cup" technique mean in GTAW?

- A) Bracing the nozzle cup against the base metal and rocking it to move the torch steadily
- B) Oscillating the filler rod in and out of the leading edge of the puddle to control bead width
- C) Moving the torch in a wide circular pattern so the heat spreads evenly across the joint and the puddle stays fluid
- D) Whipping the torch back and forth to let the puddle cool between advances

**Q23.** What is the purpose of the liner in a GMAW torch?

- A) To regulate shielding gas flow
- B) To insulate the torch from the work cable
- C) To guide the wire to the contact tip
- D) To cool the contact tip

**Q24.** What is the purpose of preheating steel before SMAW?

- A) To improve slag fluidity for easier removal
- B) To increase arc voltage
- C) To slow the cooling rate of the joint
- D) To increase the deposition rate

**Q25.** What causes "sugaring" (oxidation/discolouration) on the back side of a stainless steel GTAW root pass?

- A) Lack of back purge on the root side
- B) Excessive heat input from high amperage
- C) Insufficient post-flow gas time
- D) Using the wrong filler wire for the base metal

**Q26.** What is the effect of increasing arc length in SMAW beyond the optimal distance?

- A) Reduces heat input
- B) Increases penetration
- C) Causes porosity and spatter
- D) Improves fusion at the weld toes

**Q27.** What is "burn-back" in GMAW?

- A) Porosity from shielding gas loss
- B) The wire burning back to the contact tip
- C) Undercutting at high wire feed speeds
- D) Excessive spatter on the base metal

**Q28. What is the leg of a fillet weld?**

- A) The height of the weld reinforcement above the base metal
- B) The distance from the weld root to the toe
- C) The total length of the fillet weld deposit
- D) The total depth of penetration into the base metal

**Q29. Welding cable size is selected based on:**

- A) The amperage and duty cycle
- B) The type of electrode being used
- C) The length of the cable only
- D) The colour of the insulation

**Q30. What does WHMIS stand for?**

- A) Workshop Handling & Machinery Inspection Safety
- B) Workplace Hazardous Materials Information System
- C) Workplace Hazard Management and Inspection Sheet
- D) Welding Health and Material Inspection Standards

**Q31. What type of tungsten tip preparation is used for GTAW on steel with DCEN?**

- A) A split (notched) tip
- B) A sharp tapered point
- C) A rounded ball tip
- D) A flat-ground tip

**Q32. Why is AC polarity used for GTAW of aluminum?**

- A) AC lowers total heat input so thin aluminum will not burn through
- B) AC gives deeper penetration into aluminum than straight polarity DCEN
- C) AC prevents tungsten contamination
- D) The EP half-cycle removes the aluminum oxide layer

**Q33. Carbon equivalent (CE) formula is used to determine:**

- A) The post-weld heat treatment temperature
- B) The tensile strength of the completed weld
- C) The hardenability and preheat requirement
- D) The carbon content of the weld metal

**Q34. What is the neutral flame setting in oxyfuel welding?**

- A) Equal volumes of oxygen and acetylene
- B) Oxygen valve fully open, acetylene barely open
- C) Excess oxygen, no feather
- D) Excess acetylene, large feather present

**Q35. Welding on a container that previously held flammable liquid is safe when:**

- A) The container has been left open to air out for a full 24 hours
- B) The welder works quickly so as to minimize the heat input
- C) The container is purged, tested and certified gas free
- D) The liquid has been removed and the container rinsed with water

**Q36. What is a flashback in oxyfuel equipment?**

- A) Backfire from high acetylene flow
- B) Excess pressure causing the regulator to fail
- C) The flame going out due to low gas pressure
- D) The flame burning back into the torch or hoses

**Q37. Which shielding gas mixture is most commonly used for GMAW of mild steel?**

- A) 50% Argon / 50% Helium
- B) 100% carbon dioxide
- C) 100% Argon
- D) 75% Argon / 25% CO<sub>2</sub>

**Q38. What is hydrogen-induced cracking (HIC) and which electrode type is most susceptible?**

- A) Cracking from rapid cooling and quenching of the weld; low-hydrogen E7018 electrodes most susceptible
- B) Cracking from slag entrapment between passes; iron powder E7024 electrodes most susceptible
- C) Cracking from excessive amperage; rutile E6013 electrodes most susceptible
- D) Cracking from dissolved hydrogen in the HAZ; high-cellulosic electrodes most susceptible

**Q39. What is the gas lens in a GTAW torch?**

- A) A gas pressure regulator built into the torch body
- B) A glass lens that focuses the shielding gas into a beam
- C) A lens that focuses the arc onto a smaller area for precision
- D) A mesh screen that produces smooth, laminar gas flow

**Q40. What is the approximate spray transition current for ER70S-6 wire (0.9 mm / 0.035") with 98% argon / 2% oxygen shielding gas?**

- A) Approximately 230–250 A
- B) Approximately 80–100 A
- C) Approximately 165 A
- D) Spray transfer is possible with any gas above 200 A

**Q41. What polarity is typically used for FCAW-G?**

- A) AC (alternating current)
- B) DCEN (electrode negative)
- C) DCEP (electrode positive)
- D) DCEN for indoor, DCEP for outdoor



**Q42.** What causes a "crater crack" at the end of a SMAW weld?

- A) Rapid contraction of the pool at arc break
- B) Using DCEN instead of DCEP
- C) Moisture in the low-hydrogen electrode coating
- D) Excessive amperage throughout the weld

**Q43.** What is the difference between a groove weld and a fillet weld?

- A) Groove welds are partial penetration only; fillet welds are always full penetration
- B) Groove welds fill a prepared groove; fillet welds join surfaces at an angle
- C) Groove welds are used for lap joints; fillet welds are used for butt joints
- D) Groove welds require GTAW; fillet welds can use SMAW only

**Q44.** What do AWS welding symbols indicate?

- A) The welding process and shielding gas to use
- B) The type, size, and location of a weld on a drawing
- C) The brand of welding machine to use
- D) The electrode classification and storage requirements

**Q45.** What colour coding identifies a pure tungsten electrode (EWP)?

- A) Green
- B) Gold
- C) Red
- D) Grey

**Q46.** What does CTWD stand for in GMAW?

- A) Contact Tip Wear Detection
- B) Contact Tip to Work Distance
- C) Circuit Travel Wire Direction
- D) Current to Wire Diameter ratio

**Q47.** What type of slag system does E71T-1 flux-cored wire use?

- A) Iron powder — high deposition rate for flat/horizontal only
- B) Cellulosic slag — deep penetrating, used outdoors
- C) Basic slag — used for positional welding with DCEN
- D) Rutile slag — fluid, easily removed, all-position

**Q48.** What are the four metal transfer modes in GMAW?

- A) Short circuit, globular, spray, and pulsed spray
- B) Short circuit, dip transfer, MIG, and flux-cored
- C) Spray, pulse, plasma arc, and buried arc transfer
- D) Globular, spray, push transfer, and pull transfer

**Q49.** During OFC, the tip-to-work distance is increased. What effect does this have?

- A) Narrows the kerf and improves cut quality
- B) Reduces preheat effectiveness at the cut
- C) Has no effect on cut quality
- D) Increases preheat and reduces cut quality

**Q50.** A welder is welding in a low-lying area and notices dizziness. The most likely hazard is:

- A) Nitrogen narcosis from shielding gas
- B) Carbon monoxide from the electrode coating
- C) UV radiation causing nausea
- D) Oxygen displacement by argon or CO<sub>2</sub>

## Answer Key & Explanations

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### Q1. Answer: C

P91 (9% Cr, 1% Mo) creep-resistant steel requires matching filler metal ER90S-B9 (or equivalent). Standard carbon steel or stainless filler would produce mismatched mechanical properties.

### Q2. Answer: B

The feed path controls how smoothly the wire reaches the arc. Worn or wrongly grooved drive rolls, incorrect drive tension, a kinked or clogged liner, a worn contact tip or a tangled spool make the wire arrive in surges, so the arc lengthens and shortens, pops, and burns back. The other faults leave the feed alone and show themselves differently: low shielding gas flow shows up as porosity, too slow a travel speed as excessive buildup and cold lap, and a wrong work angle as undercut and poor bead placement. Check the feed system first — liner condition and length, drive roll size and groove type for the wire being run, drive tension, and the contact tip.

### Q3. Answer: B

E6010 has a high-cellulosic sodium coating that produces a forceful, deep-penetrating arc. It requires DCEP and is ideal for root passes on pipe and dirty/rusty steel.

### Q4. Answer: B

A stringer bead is a narrow, straight pass with no side-to-side weaving motion. It provides better mechanical properties (especially toughness) than a wide weave bead.

### Q5. Answer: B

Damaged insulation creates electrocution and fire hazard. Equipment must be repaired or replaced before use.

### Q6. Answer: C

The HAZ is the region of base metal adjacent to the fusion zone that has not melted but experienced temperatures high enough to alter its microstructure and properties (grain growth, hardening).

### Q7. Answer: D

Anti-spatter spray is applied to nozzles and workpiece surfaces to prevent spatter from adhering and bonding. It reduces nozzle cleaning frequency and cleanup time.

**Q8. Answer: C**

E6013 (rutile-coated) produces a soft arc, light spatter, easy slag removal, and shallow penetration — making it ideal for thin sheet metal where burn-through is the primary risk. Wrong answers: E6010 and E6011 have deep penetration and forceful arc (designed for root passes and pipe), which would easily burn through thin material. E7018 (low-hydrogen) is excellent for structural work but requires an oven and produces a broader bead; it is not the preferred choice for light gauge sheet. E6013's easy restriking and smooth bead also make it a good learning electrode, but its low penetration means it's NOT suitable for code-quality root passes or high-strength joints where full fusion is required.

**Q9. Answer: C**

Low arc voltage relative to wire feed speed causes an unstable arc and excessive spatter in FCAW-G. Proper voltage-WFS balance is critical. Also, 100% CO<sub>2</sub> produces more spatter than Ar/CO<sub>2</sub> mixes.

**Q10. Answer: D**

Rank the modes by arc length and you rank them by voltage. Short circuit sits at the bottom: the arc is so short that the wire dips into the puddle and extinguishes the arc many times a second. Globular sits in the middle, with a longer arc carrying drops larger than the wire diameter across it. True axial spray sits at the top, because the arc has to be long enough and hot enough to pinch the wire tip into a fine stream of droplets smaller than the wire. Pulsed spray is the trap here: its peak current does climb above the spray transition, but the whole point of pulsing is to get spray-type droplet transfer at a lower average voltage and current, which is exactly what lets it run out of position. Voltage alone is not enough either — true spray also needs an argon-rich shielding gas, and a CO<sub>2</sub>-rich gas will not give axial spray at any voltage setting.

**Q11. Answer: D**

Read E71T-1 one character at a time. E is electrode. 7 is the minimum tensile strength, 70 000 psi. Do not convert that number yourself: AWS publishes a separate metric classification, and in it the same wire is E491T-1, where 49 stands for 490 MPa. AWS A5.20/A5.20M:2005 Table 1 requires 70–95 ksi for the E7XT-1 classifications and Table 1M requires 490–670 MPa for E49XT-1, and the specification states that the two systems 'are not equivalent, each system must be used independently of the other.' The 1 that sits ahead of the T is the position digit and means all-position. T means the wire is tubular, that is flux-cored, as opposed to a solid GMAW wire. The -1 that follows the T is the usability designator: it identifies a rutile-type, fast-freezing slag system run on DCEP with CO<sub>2</sub> or an argon/CO<sub>2</sub> mix as the shielding gas. The two 1s in the designation therefore carry different meanings, and it is the trailing one that describes gas and polarity. Also keep the shielding split straight: E71T-1 requires an external shielding gas (FCAW-G), while a self-shielded wire such as E71T-11 or E71T-8 generates its own shielding from the core (FCAW-S) and runs with no gas bottle at all.

**Q12. Answer: C**

In Canada welding ventilation is an exposure-limit duty, not a fixed airflow number. Provincial regulation makes CSA W117.2 (Safety in welding, cutting, and allied processes) mandatory and then requires capture at source: British Columbia's OHS Regulation section 12.112 requires welding, cutting and similar processes to be carried out according to CSA W117.2, and section 12.114 requires effective local exhaust ventilation at any fixed work station to minimize worker exposure to harmful air contaminants. The test of adequacy is whether fume and gas stay below the occupational exposure limits for the metals and coatings involved — use the edition of W117.2 and the exposure limits your province adopts. The 2,000 cfm per welder figure offered here is American, not Canadian: OSHA 1910.252 sets that rate where the space is less than 10,000 cubic feet per welder or the ceiling is less than 16 feet, and it has no force here. The 1,000 cfm variant simply invents a Canadian version of that same US ceiling-height trigger. The idea that ventilation only matters above some current is false — fume is produced at every setting, and there is no amperage below which ventilation stops being required. The one Canadian airflow figure worth memorizing comes from the Canadian Centre for Occupational Health and Safety: a moveable exhaust hood should provide an air velocity of at least 100 ft/min (0.5 m/s) across the welding arc, placed as close to the work as practical. Local exhaust is mandatory regardless of general airflow when welding coated, galvanized or stainless material, where zinc oxide and hexavalent chromium exceed exposure limits even in a well-ventilated shop. AWS D1.1 and CSA W59 are fabrication codes and set no ventilation requirement.

**Q13. Answer: A**

The Charpy test strikes a notched specimen with a pendulum and measures the energy absorbed before fracture (in joules). It evaluates toughness at specified temperatures, critical for pressure vessels and low-temperature service.

**Q14. Answer: A**

OFC works by preheating steel to kindling temperature (~870°C/1600°F), then directing a high-pressure oxygen jet at the preheated metal to oxidize (burn) it rapidly.

**Q15. Answer: C**

Silicon (Si) and manganese (Mn) are deoxidizers. ER70S-6's higher Si/Mn content scavenges oxygen from the weld pool, improving weld quality on slightly rusty or dirty base metal and giving better wetting.

**Q16. Answer: C**

When travel speed is too fast, the oxygen jet cannot cut through the full thickness vertically, causing the cut to lag behind and produce curved drag lines. Travel that is too slow does the opposite: the top edge melts and rounds and slag clings underneath, but the cut face stays square, so no drag develops. Excessive cutting oxygen pressure gives a wide, rough, gouged kerf rather than drag, and oversized preheat flames melt and round the top edge.

**Q17. Answer: A**

Globular transfer occurs at intermediate voltages between short circuit and spray thresholds. Large irregular droplets — bigger than the wire diameter — form and fall into the pool, causing significant spatter. Generally avoided in production.

**Q18. Answer: C**

These are worm tracks, also called gas tracks: shallow tunnels or grooves in the surface of the bead, hidden under the slag until it comes off. The flux core generates gas as it burns, and most of that gas escapes up through the puddle. When the slag blanket freezes over the top before the last of it is out, the gas is trapped at the interface and leaves its path printed along the crown. That makes worm tracks a surface signature, quite unlike the fine scattered voids through the body of the bead that come from a damp consumable. Too long an arc is the usual reason — it gives the core more gas to shed and less time in which to shed it — and wire that has picked up moisture adds to the gas load, so shorten the arc, slow down enough to let the puddle degas, and use wire that has been stored dry. The other three answers name real flux-cored defects with different signatures: undercut is a groove in the base metal at the toe, a slag inclusion is non-metallic material buried inside the deposit, and an arc strike is a mark outside the weld itself.

**Q19. Answer: C**

DCEP (reverse polarity) means the electrode is connected to the positive terminal and the work to the negative terminal. In SMAW this is the deeper-penetrating polarity, and it is what E6010 and E7018 are run on. Do not carry that rule across to GTAW: the tungsten there is not consumed, and DCEN is the polarity that concentrates the heat in the work and gives the deeper penetration. The rule about which polarity penetrates belongs to the process, not to the letters.

**Q20. Answer: B**

HF start initiates the arc via a high-frequency spark across the gap between the electrode and workpiece — no contact required. This prevents tungsten contamination and electrode tip damage.

**Q21. Answer: A**

Low-hydrogen electrodes absorb moisture rapidly. Storage in a rod oven at 120–150°C keeps the coating dry, and a dry coating is what keeps diffusible hydrogen out of the weld metal — moisture picked up by the flux is the source of the hydrogen behind hydrogen-induced (cold) cracking and porosity.

**Q22. Answer: A**

"Walking the cup" means bracing the ceramic nozzle against the pipe and using a rocking or rolling motion to move the torch. This provides a stable torch angle and consistent arc length for pipe welding.

**Q23. Answer: C**

The liner (conduit liner) guides the wire electrode from the drive rolls through the torch body to the contact tip, preventing kinking and ensuring smooth wire feed.

**Q24. Answer: C**

Preheat slows the cooling rate of the HAZ, reducing hardness and the risk of hydrogen-induced cracking, especially in high-carbon or low-alloy steels.

**Q25. Answer: A**

"Sugaring" is a rough, oxidized surface on the weld root caused by atmospheric oxygen reacting with chromium in the hot metal, forming chromium oxides ( $\text{Cr}_2\text{O}_3$ ) and depleting the corrosion-resistant chromium from the surface.

**Q26. Answer: C**

Longer arc length increases arc voltage and heat, but also exposes the molten weld pool to atmospheric contamination, causing porosity and spatter. Proper arc length equals electrode diameter.

**Q27. Answer: B**

Burn-back occurs when the wire electrode burns back and fuses to the contact tip, stopping the weld. Caused by low wire feed speed relative to voltage.

**Q28. Answer: B**

The leg of a fillet weld is the distance from the root of the joint to the toe of the weld, measured along the fusion face. For an equal-leg fillet, both legs are the same length.

**Q29. Answer: A**

Cable must handle the amperage and duty cycle of the welding process without overheating. Undersized cable creates excessive resistance, heat, and fire hazard.

**Q30. Answer: B**

WHMIS (Workplace Hazardous Materials Information System) is Canada's national hazard communication standard for hazardous products.

**Q31. Answer: B**

For DCEN GTAW (steel, stainless, titanium), the tungsten is ground to a tapered point. For AC GTAW (aluminum), a rounded ball forms naturally at the tip.

**Q32. Answer: D**

Aluminum has a tenacious oxide layer ( $\text{Al}_2\text{O}_3$ , melting point  $2050^\circ\text{C}$ ) that prevents fusion. During the electrode positive (DCEP) half of AC, electrons flow from the work to the electrode, breaking up the oxide — this is cathodic cleaning.

**Q33. Answer: C**

$\text{CE} = \text{C} + \text{Mn}/6 + (\text{Cr} + \text{Mo} + \text{V})/5 + (\text{Ni} + \text{Cu})/15$ . Higher CE means greater hardenability of the base metal, requiring preheat to slow cooling and reduce HIC risk.

**Q34. Answer: A**

A neutral flame has a 1:1 oxygen-to-acetylene ratio, producing a sharp bright inner cone with no feather — suitable for most welding.

**Q35. Answer: C**

Containers must be thoroughly cleaned, purged, and then gas-tested and certified free of flammable vapour before any hot work — explosive residues can remain long after airing, soaked into scale, seams and sludge. Where a residual reading is tolerated at all, the hot-work limit is far tighter than the limit for cold work: Ontario's confined space guideline permits cold work under 10% LEL but hot work only under 5% LEL, and British Columbia requires flammable gases and vapours to be held below 20% of the LEL. A rinse, or a day with the lid off, proves nothing.

**Q36. Answer: D**

Flashback occurs when the flame burns back into the hoses or torch body, causing a hissing/squealing sound. It can cause hose fires or cylinder explosions.

**Q37. Answer: D**

C25 (75% Ar / 25% CO<sub>2</sub>) is the most widely used mixture for mild steel GMAW. It provides a stable arc, good penetration, and lower spatter than 100% CO<sub>2</sub>.

**Q38. Answer: D**

HIC (cold cracking) occurs when hydrogen from the electrode diffuses into the HAZ and combines with residual stress and microstructure to cause delayed cracking. Cellulosic (E6010/E6011) electrodes produce the most hydrogen.

**Q39. Answer: D**

A gas lens replaces the standard collet body. Its porous metal mesh screen produces a smooth, laminar (non-turbulent) flow of shielding gas, extending coverage and enabling longer electrode stickout.

**Q40. Answer: C**

Spray transfer requires an argon-rich gas (at least ~80% Ar). With 98Ar/2O<sub>2</sub>, 0.9 mm ER70S-6 transitions from globular to spray at roughly 165 A; with 90Ar/10CO<sub>2</sub> the transition rises to about 200 A. With C25 (75/25) true axial spray cannot be achieved — the high CO<sub>2</sub> content disrupts the arc column.



**Q41. Answer: C**

Gas-shielded flux-cored wire is run on DCEP, direct current electrode positive, the same polarity as GMAW solid wire. AWS A5.20/A5.20M:2005 Table 2, Electrode Usability Requirements, defines the term in a footnote as direct current electrode positive, or reverse polarity, and its polarity column is where the word "typically" in the question comes from. The gas-shielded usability designators are 1, 2, 5, 9 and 12. Designators 1, 2, 9 and 12, which include the widely used E71T-1, are listed for DCEP, and the annex describes the T-1 wires as designed for single and multiple pass welding using DCEP. Designator 5 is the exception: the flat-and-horizontal E70T-5C and E70T-5M electrodes are listed DCEP, but the all-position E71T-5C and E71T-5M electrodes are listed "DCEP or DCEN", with a footnote that some of them are recommended for DCEN to improve out-of-position welding and that the manufacturer should be consulted. Self-shielded wire is where polarity really varies: designators 3, 4 and 6 are DCEP, while 7, 8, 10, 11, 13 and 14, the majority and the common E71T-8 and E71T-11 among them, are DCEN. So a gas-shielded wire is set up electrode positive unless its classification or the manufacturer's data sheet says otherwise, and the leads are never left where the last spool had them: wrong polarity on a flux-cored wire shows up as an erratic arc, heavy spatter, porosity and poor fusion. The polarity column of that table carries only DCEP, DCEN, or "not specified" for the G classifications, so AC is not a classified polarity for any carbon steel flux-cored designator. Nor does any designator tie polarity to where the work is done: indoors or outdoors decides between gas-shielded and self-shielded wire, not between electrode positive and electrode negative.

**Q42. Answer: A**

When the arc is suddenly extinguished, the weld pool shrinks rapidly. If the crater is not filled, tensile stresses from contraction cause a star-shaped crater crack.

**Q43. Answer: B**

A groove weld occupies a groove or gap prepared in the base metal (butt joint). A fillet weld is a triangular cross-section weld joining surfaces at approximately 90° (T-joint, lap joint, corner joint).

**Q44. Answer: B**

AWS A2.4, Standard Symbols for Welding, Brazing, and Nondestructive Examination, is the welding-symbol reference used on Canadian drawings. On structural work it reaches you through CSA W59, Welded steel construction, the Canadian structural welding standard, which adopts the AWS A2.4 symbols together with additional conventions of its own — so a Canadian shop drawing is read to AWS A2.4 alongside CSA W59. Welding symbols communicate weld type, size, length, location and finishing requirements on engineering drawings without written notes. Key elements: the reference line is horizontal — a symbol below the line means an arrow side weld, a symbol above means an other side weld, and symbols on both sides mean weld both sides. The tail carries the welding process or a specification reference, but the symbol does not name the shielding gas and it never names a machine. Common Red Seal exam traps: a solid flag (pennant) at the junction of the arrow and the reference line means field weld, while a circle at that same junction means weld all around — two different symbols in the same place, so do not read the flag as the circle; and the size is placed to the left of the weld symbol with length and pitch to its right, while contour and finish symbols (G = grinding, C = chipping, M = machining) are placed beyond the weld symbol, the finish symbol sitting outside the contour symbol away from the reference line. Wrong answers confuse 'arrow side' with 'front of the part' — arrow side means the side of the joint the arrow touches, whatever the orientation of the part.

**Q45. Answer: A**

AWS A5.12M/A5.12 (a modified adoption of ISO 6848) colour coding: pure tungsten, classification EWP, is Green. EWP is unalloyed tungsten, 99.5% minimum, with no oxide addition. It is used with alternating current for welding aluminum and magnesium, where it holds a clean balled end. The other three colours in this question belong to different electrodes: Grey is 2% ceriated (EWCe-2), Red is 2% thoriated (EWTh-2), and Gold is 1.5% lanthanated (EWLa-1.5). One change catches people out on the exam and in the rod drawer: the 2009 edition of A5.12 moved EWCe-2 from orange to grey, and grey is no longer the EWG code. Older charts and older stock still show orange for ceriated, so trust the current colour band, not the wall chart.

**Q46. Answer: B**

CTWD (Contact Tip to Work Distance) is the distance from the contact tip to the workpiece. It affects electrical stickout and therefore current/deposition rate.

**Q47. Answer: D**

E71T-1 is a rutile-type flux-cored wire used with CO<sub>2</sub> or Ar/CO<sub>2</sub> shielding gas. It produces a fluid, easily removed slag and is suitable for all-position welding.

**Q48. Answer: A**

The four GMAW transfer modes: (1) Short-circuit transfer — the lowest arc voltage of the four, the wire periodically touches the puddle and extinguishes the arc, suitable for thin gauge and out-of-position, but a cold lap and incomplete fusion risk at heavy thickness; (2) Globular transfer — a longer arc carrying drops larger than the wire diameter, transferring erratically with high spatter, generally not a preferred mode in production; (3) Spray transfer — above the spray transition current, metal transfers as fine axial droplets, high quality and deposition rate, but restricted to flat and horizontal positions and requiring an argon-rich gas; (4) Pulsed spray — a background current below the spray transition and a peak pulse current above it, which gives spray-type droplet transfer at a lower average current and heat input. Red Seal key point: spray transfer cannot be used out of position because gravity makes the puddle sag, and pulsed spray is what solves that.

**Q49. Answer: B**

Increasing tip-to-work distance reduces heat intensity at the workpiece, potentially insufficient for proper preheat and causing poor cutting or failure to cut.

**Q50. Answer: D**

Argon and CO<sub>2</sub> are heavier than air and can accumulate at floor level in low-lying areas, displacing oxygen and causing asphyxiation.

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